THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 89

COLLAPSIBLE GEODESIC HOBERMAN ROLL-BALLS & S-BEND SCOOP VALVES

Packs to 50cm — tumbles alien hillsides wide

the S-bend is a valve with no moving parts

geometry does the machine's job

SCISSOR-BRACE FRAME · GRAVITY CHECK-VALVES

DROP THE BALL — LET THE HILL WORK

Passive survey harvester — v1.0 in force

GP-IM-089

Revision v1.0 — 22 August 2026

90 of 290

VETTED — 4-AI WEB — BATCH 15 — PASS

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THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 89

PHASE 89: COLLAPSIBLE GEODESIC HOBERMAN ROLL-BALLS & S-BEND SUGAR-SCOOP VALVES — STATUS: CURRENT. A survey harvester that packs to a 50cm cube and deploys as a tumbling geodesic ball. Phase 89 wraps an expandable scissor-brace Hoberman frame in peripheral sugar-scoop intakes; material scraped during rotational travel slips down hollow S-bend conduits under gravity alone — a check-valve with no moving parts, no motor, no seal to fail. Crane-dropped onto unmapped alien hillsides, the ball expands, tumbles, and harvests passively; back in the bay it compresses, takes the Phase 12 steam-and-chemical pre-wash, and is emptied by suited crew inside a sealed isolation box. The workshop toy analog rides in the masthead, verbatim: the industrial water-wheel sugar scoop and the expandable geodesic Hoberman ball.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-089, v1.0 — 22 August 2026. The ninetieth manual of the program — 90 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the roll-ball law as seated (contents line, the three design bodies — verbatim) — §2 why it works (gravity valves and scissor geometry — graded) — §3 the system in the ball — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 89: **Collapsible Geodesic Hoberman Roll-Balls & S-Bend Sugar-Scoop Valves.** Designs rapid-deployment storage hoppers using high-strength expanding geodesic linkages, regulated via manual, un-complicated mechanical S-bend "sugar-scoop" valves to eliminate material bridging and jamming during cargo drops.'

THE MASTHEAD LINE (verbatim): 'Workshop Toy Analog: The Industrial Water-Wheel Sugar Scoop & Expandable Geodesic Hoberman Ball [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Passive Gravitational Multi-Depth Soil Sampling & High-Secrecy Biocontainment *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']* [extracted verbatim from the combined masthead line]'

THE S-BEND PERIPHERAL SUGAR-SCOOP VALVES (VERBATIM)

* Intake Geometry: Peripheral shell surfaces are lined with high-durability, low-profile kitchen-grade sugar scoops. * The Gravity Check-Valve: Connects the scoop body to an internal collection chamber via a hollow S-bend conduit. * Non-Mechanical Capture: Material scraped during rotational travel slips down the S-bend neck under gravity. The curved geometry blocks backward fluid/solid migration during inverted wheel tracks, permanently trapping the micro-sample without motorized check valves.

THE COLLAPSIBLE GEODESIC VORTEX BALL (VERBATIM)

* Volumetric Optimization: Harvester chassis utilizes an expandable scissor-brace geodesic architecture, compressing to a dense 50cm baseline to minimize lander bay cargo footprints. * Passive Gravitational Survey: Deployed via crane vectors onto unmapped alien hillsides. The ball expands to a wide radius and tumbles down angular slopes via pure gravity, bouncing over terrain anomalies to execute rapid, multi-point ground-material capture.

STAGED BIOCONTAINMENT QUARANTINE LOOP (VERBATIM)

* Core Pre-Wash Sequence: Post-retrieval, the compressed 50cm ball undergoes an automated Phase 12 steam and chemical pre-wash inside the landing bay to incinerate surface-level pathogens. * Manual Extractions Interface: Suited personnel manually empty the internal chambers onto localized bio-plate collection boxes within a sealed isolation enclosure. * Parallel Decontamination Tracks: Sealed bio-boxes are routed directly to Phase 24 laboratory scanning grids. The empty harvester drone is transferred to a high-intensity hardware decontamination core, completely isolating the crew atmosphere from alien toxicity.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE S-BEND CHECK-VALVE, AFFIRMED: a hollow S-bend conduit is a one-way geometry enforced by gravity and shape — material slides down the neck and cannot climb back. No actuator, no seal, no power: the valve is the curve. This is the same class as the Phase 84 figure-8 and the Phase 95 lever-clinch — geometry doing the work a mechanism would otherwise owe maintenance for.

THE HOBERMAN ARCHITECTURE, AFFIRMED: scissor-brace geodesic expansion is proven mechanics from toy to antenna — a dense packed baseline, a wide deployed radius, and every strut sharing the load. Packing a survey tool to 50cm and deploying it to a hillside-wide tumbler is exactly what the geometry is for.

THE PASSIVE SURVEY, AFFIRMED: tumbling under gravity down angular slopes costs no ship power and needs no drivetrain; the terrain does the propulsion (Phase 90 carries the powered end of that family — flat ground, where gravity quits).

THE QUARANTINE LOOP, AFFIRMED AS DISCIPLINE: steam-and-chemical pre-wash on the compressed ball before any manual extraction inside sealed isolation is the correct order of operations — sterilize the outside before you open the inside.

§3 — THE SYSTEM IN THE BALL

- THE FRAME: expandable scissor-brace geodesic — 50cm packed baseline, wide-radius deployed tumble.
- THE SCOOPS: peripheral high-durability sugar-scoop intakes lining the shell — low-profile, kitchen-grade logic at frontier scale.
- THE VALVES: hollow S-bend conduits to the internal collection chamber — gravity check-valves, non-mechanical capture.
- THE WASH: Phase 12 steam-and-chemical pre-wash in the landing bay, ball compressed, surface pathogens incinerated.
- THE TABLE: suited manual extraction onto bio-plate collection boxes inside sealed isolation.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE GEOMETRY-IS-THE-VALVE DOCTRINE (seated with the S-bends): where gravity points, shape can do a machine's job — the curve never wears out.
- THE TOY-TO-TOOL DOCTRINE (the masthead analog, verbatim): the Hoberman ball and the water-wheel sugar scoop — proven toy mechanics, grown into survey hardware.
- THE PACK-SMALL DOCTRINE (seated with the frame): lander-bay volume is the expensive volume — tools that compress to a tenth of their working size earn their berth.
- THE STERILIZE-FIRST DOCTRINE (seated with the quarantine loop): wash the outside before opening the inside — the order is the safety.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 15, SEATED — the verdict: ' Mechanical concept PASS. The ball expands from a compact configuration, rolls/tumbles under gravity, collects material through peripheral scoops, and uses an S-bend [conduit as a gravity check-valve].' The batch-15 cross-check (sitting 268) graded the run green and records no flags on 89. The audit affirms: the mechanical concept is clean; the owed items are bench numbers, not physics (see §9). Graded under the 4-AI vet web (sitting 565).

§6 — BUILD SEQUENCE

- STEP 1 — Build the frame: scissor-brace geodesic, 50cm packed; expansion/contraction cycle life benched.
- STEP 2 — Line the scoops: peripheral scoop array fitted; scoop edge wear logged per terrain class.
- STEP 3 — Prove the valves: S-bend flow bench — capture rate, clog points, clear-back procedure; the valve must fail safe, not fail shut.
- STEP 4 — Drop the ball: crane-deploy tumble trials on graded slopes; harvest mass per run logged.
- STEP 5 — Run the wash: Phase 12 pre-wash cycle on the compressed ball; pathogen kill assay before extraction.
- STEP 6 — Publish the ledger: cycle life, wear, capture rates, wash efficacy — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 90 — the eccentric-weight motors (GP-IM-090): the powered sibling — the same frame, driven from inside when the ground goes flat.
- PHASE 12 — the bio-contamination family (manual owed): the steam-and-chemical pre-wash.
- PHASE 84 — the figure-8 doctrine (GP-IM-084): geometry-as-machine, the same class as the S-bend valve.
- PHASE 49 — the humanoid units (GP-IM-049): extraction duty where the terrain or the quarantine excludes crew.

§8 — DO-NOT-BUILD

- NO motorized valves on the scoops — the S-bend is the valve; adding actuators adds failure points to a solved problem.
- NEVER skip the pre-wash — the wash comes before the opening, every time, no matter how clean the hillside looked.
- NEVER deploy under power on slopes — the tumble is passive by design; driven rolling on unknown terrain is how survey tools become wreckage.
- NO fixed-frame substitutes — the 50cm packed baseline is the berth cost that makes the tool shippable.

§9 — OWED

THE BENCH NUMBERS, OWED: scoop edge-wear ledger per terrain class; S-bend capture/clog/clear rates; expansion cycle life; wash-efficacy assay per pre-wash cycle.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-089, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The ninetieth manual of the program — 90 of 290. Built 22 August 2026, second of the 20-manual 'clear the 100 mark' shift ('do another 20 while i away so we clear the 100 mark for the day when i get back'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564 and 566): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back'. The phase's own chain, all seated in the master: the contents line and the three design bodies (as seated — the workshop toy analog in the masthead, verbatim); the 12 August naming batch (formerly 'GEODESIC ROLL-BALL SOIL HARVESTER & BIO-AUDIT MATRIX'); the vet batch 15 grade in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the roll-ball bench ledger, or his word.

END OF MANUAL — PHASE 89